

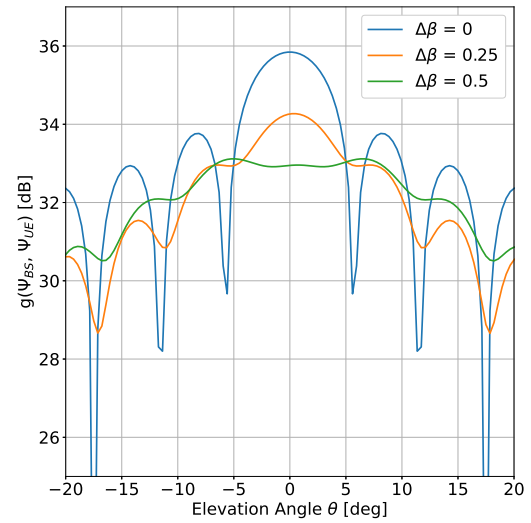
## Research Project

### IRS Codebook Design under Hardware Considerations

for

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Intelligent reflecting surfaces (IRSs) can provide an additional propagation path in wireless communication systems. This is especially useful for wireless communication systems employing high carrier frequencies if the Line-of-Sight (LoS) from the transmitter to the receiver suffers from severe attenuation from large objects, such as buildings. IRS comprise a large number of unit cells that can apply a phase shift to the reflected electromagnetic waves. The superposition of the reflections from all unit cells leads to reflection behaviors depending on the phases. Early works on IRS focused on optimizing the phase shifts for perfect channel state information (CSI). Later, however, the problem of channel acquisition was commonly acknowledged and its necessarily large overhead makes such approaches challenging. To avoid resource-intensive channel estimation, the IRS can be configured by selecting predefined codewords from a codebook, which can be done based on measurements of the user's direction. An example for the beam pattern of three different codewords targeting the same direction, but with different beamwidth is shown in Figure 1. In this project, a codebook that is suitable for the near-field shall be derived. In the practical implementation of IRS hardware, the codebooks are limited by the IRS technology, e.g., copper patches adjusted by PIN diodes. Therefore, the limitations and inaccuracies of the hardware shall be considered.



**Figure 1:** Example for IRS reflection patterns. Here, the parameter  $\Delta\beta$  changes the width of the beam.

The main guidelines for the project are as follows:

- Literature review on IRS models and codebook designs.
- Comparison of existing codebooks under hardware impairments and near-field assumption.
- Design of a robust codebook suitable for the near-field under hardware impairments.

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| PREREQUISITES      | Knowledge of digital communication and mathematical optimization. |
| PROGRAMMING SKILLS | Experience in Python or MATLAB.                                   |

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